Data Quality Assessment — Dataset 2 (Plant-Level Solar, 2 Plants × 22 Inverters)

Assessed 01/09/2026. 4 source sheets, 142,917 data rows. Every figure below was measured across ALL rows, not sampled.

1. Is it better than the previous dataset?

Short answer: it is a much cleaner dataset but a narrower one. The cleaning burden drops by roughly 80%; the analytical scope drops just as sharply. Detail below.

#	Dimension	Previous dataset (42 sites / 5 campuses)	This dataset (2 plants / 44 inverters)	Verdict
1	Missing / blank values	599,377 blank generation cells (57.2%) plus 270,003 blank weather cells (28.8% temps, 43.8% wind).	Zero blanks. Not one, across 136,476 generation rows × 4 measures and 6,441 weather rows × 3 measures.	**MUCH BETTER**
2	Irradiance measurement	None. Temperature and humidity were only proxies for sunshine, so a true Performance Ratio was impossible.	IRRADIATION present on every weather row: range 0 – 1.2217, zero blanks, zero negatives.	**TRANSFORMATIVE**
3	Module (panel) temperature	None. Only ambient air temperature.	MODULE_TEMPERATURE present: 18.1 – 66.6°C vs ambient 20.4 – 39.2°C. Enables thermal-derating analysis.	**NEW CAPABILITY**
4	AC vs DC power	A single generation column. Inverter conversion loss was invisible.	Both DC_POWER and AC_POWER on every row → inverter efficiency is directly measurable per inverter.	**NEW CAPABILITY**
5	Row-limit truncation	Critical: 61% of rows silently lost at Excel's 1,048,576-row ceiling. Needed a Power Query re-import to recover.	None possible. Largest sheet is 68,778 rows – 6.6% of the ceiling.	**MUCH BETTER**
6	Duplicates & timestamp hygiene	No duplicates, but generation and weather sat on slightly offset grids.	No duplicate (inverter, timestamp) pairs. 100% of timestamps land exactly on the :00/:15/:30/:45 grid. Already stored as datetime serials, not text.	**BETTER**
7	Interval coverage	77% average per site; worst site 24%.	Plant 1: 95.1–96.7% per inverter. Plant 2: 72.2–99.8% (three inverters at 2,355 of 3,264).	**BETTER**
8	Night-time handling	Blank cells – ambiguous between 'no output' and 'no reading'.	Explicit 0 values (46.5% of P1 rows, 52.7% of P2). Unambiguous, and AVERAGE behaves predictably.	**BETTER**
9	Time span	844 days – 01/01/2020 to 23/04/2022 (2.3 years).	34 days – 15/05/2020 to 17/06/2020. That is the entire dataset.	**MUCH WORSE**
10	Seasonality & year-on-year	Two full seasonal cycles plus a genuine YoY comparison (2021 vs 2020).	Impossible. 34 days inside a single season. No YoY, no seasonal profile, no time intelligence. A date dimension is still needed, but SAMEPERIODLASTYEAR and every seasonal visual are dead.	**CRITICAL LOSS**
11	Asset / capacity register	kWp for 25 of 42 sites, plus panel model, panel count, inverter and optimiser hardware, and lat/long.	None at all. No kWp, no panel data, no hardware, no location, no commissioning date. No dimension table exists in the source – only two fact tables and two sensor tables.	**WORSE**
12	Entity naming	SiteKey 1–42 with campus names – human readable.	SOURCE_KEY is an opaque 15-character hash ('1BY6WEcLGh8j5v7'). Unusable on an axis or in a slicer.	**WORSE – fixed**
13	Internal consistency	Strong: panels × watts reconciled to reported kWp at 0.0% variance on all 25 testable sites.	Two real defects found: Plant 1 DC_POWER is inflated 10×, and 71.6% of Plant 2 TOTAL_YIELD values are on the wrong scale. Both are diagnosed and both are fixable – see issues G1 and G2.	**NEW ISSUES**
14	Grain alignment (weather vs output)	Weather at campus grain vs generation at site grain – 27 sites shared one weather record.	Weather at plant grain vs generation at inverter grain – 22 inverters share one sensor. Same class of mismatch, but only 2 levels and the sensor sits physically on site.	**SIMILAR**

Bottom line

Better data, smaller question.	You were right to switch. The previous dataset needed 13 separate cleaning interventions and still left the headline KPI covering only 60% of the fleet. This one needs 2 real fixes and then it is genuinely clean — and	**SWITCH JUSTIFIED**

	irradiance means you can finally build a true Performance Ratio, which is the metric the whole project was reaching for and could never have.	
But redesign the dashboards.	34 days kills Dashboard 1's seasonal and YoY visuals and Dashboard 3's seasonal comparison outright. The three perspectives still hold as concepts, but roughly half the specific visuals in Analysis 1 and Analysis 3 must be replaced. Section 6 sets out what survives, what dies, and what replaces it.	**ACTION NEEDED**

2. Table inventory

Source sheet	Grain (unique key)	Entities present	Data rows (live)	Role in the data model
Plant_1_Generation_Data	SOURCE_KEY + DATE_TIME (verified unique, 0 duplicates)	PLANT_ID 4135001; 22 inverters	68,778	Fact — append with Plant 2 into a single Fact_Generation
Plant_2_Generation_Data	SOURCE_KEY + DATE_TIME (verified unique, 0 duplicates)	PLANT_ID 4136001; 22 inverters	67,698	Fact — append with Plant 1
Plant_1_Weather_Sensor_Data	DATE_TIME (single sensor, so timestamp alone is the key)	1 sensor: HmiyD2TTLFNqkNe	3,182	Fact — append with Plant 2 into a single Fact_Weather
Plant_2_Weather_Sensor_Data	DATE_TIME (single sensor)	1 sensor: iq8k7ZNt4Mwm3w0	3,259	Fact — append with Plant 1
TOTAL	— no dimension table exists in the source —	2 plants, 44 inverters, 2 sensors, 34 days	**142,917**	4 facts, 0 dimensions → all 4 dimensions had to be synthesised (Section 4)

3. Issue log

ID	Where	Finding (measured across all rows)	Why it matters	Severity	Action
G1	Plant 1 generation	DC_POWER is inflated by exactly 10×. The AC/DC ratio is 9.78% for all 22 Plant 1 inverters — min and max both 9.78%, i.e. perfectly uniform — while Plant 2's 22 inverters sit at 97.76–97.92%. ×10 gives 97.8%, an exact match to Plant 2. No real inverter is 9.8% efficient.	Every DC-based metric, and any comparison of the two plants, is wrong by an order of magnitude. The perfect uniformity proves it is a systematic unit/scaling error, not corruption — which is why it is safe to correct.	**CRITICAL**	Correct in a DAX calculated column, not by editing the source: DC_Power_Corrected = IF(PLANT_ID=4135001, DC_POWER/10, DC_POWER). Use that column everywhere; never DC_POWER raw.
G2	Plant 2 generation	TOTAL_YIELD is on the wrong scale for most rows: 48,491 of 67,698 (71.6%) exceed 100,000,000, peaking at 2,247,916,295. Plant 1's entire range is 6,183,645 – 7,846,821. A further 563 Plant 2 rows are exactly 0 in what should be a monotonic cumulative meter.	Not a handful of outliers — it is the majority of the column, so it cannot be cleaned by trimming. TOTAL_YIELD is unusable for anything cross-plant.	**HIGH**	Exclude TOTAL_YIELD from the model, or use it only within a single plant. Derive all energy figures from AC_POWER instead — clean on both plants.
G3	Both generation sheets	DAILY_YIELD is a cumulative running total within each day that resets at midnight — not a per-interval value. Max 9,163 (P1) and 9,873 (P2).	Summing it inflates the answer roughly 96-fold, because every interval already contains the day's running total.	**MEDIUM**	Never sum DAILY_YIELD. Use MAX per inverter per day, or derive from AC_POWER. Set its default aggregation to MAX so a careless drag cannot sum it.
G4	Both generation sheets	Missing intervals. Expected 3,264 per inverter (34 days × 96). Plant 1 runs 3,104–3,155 (95.1–96.7%). Plant 2 runs 2,355–3,259 — three inverters sit at just 2,355 (72.2%).	Raw AC_POWER totals unfairly penalise inverters that simply logged fewer intervals. Three Plant 2 inverters are missing over a quarter of the period.	**MEDIUM**	Dim_Inverter carries LoggedIntervals and Coverage% per inverter. Normalise per logged interval and show Coverage% on every inverter ranking.
G5	Both generation sheets	DC_POWER and AC_POWER are exactly 0 on 31,951 Plant 1 rows (46.5%) and 35,662 Plant 2 rows (52.7%).	These are night intervals, recorded as explicit zeros rather than blanks — the correct, unambiguous way to do it.	**LOW - by design**	No cleaning needed. But any 'efficiency while generating' measure must filter to IRRADIATION > 0, or roughly half the rows drag the average toward zero.
W1	Both weather sheets	Missing intervals: Plant 1 has 3,182 of 3,264 (97.5%), Plant 2 has 3,259 of 3,264 (99.8%). Zero blanks within the rows that exist.	A small number of intervals have generation but no matching weather, so Performance Ratio is undefined for those.	**LOW**	Leave as gaps — do not interpolate. Dim_DateTime makes the 82 and 5 missing intervals visible rather than silent.
W2	Weather vs generation	One weather sensor per plant. All 22 inverters at a plant share a single irradiation and temperature reading.	Weather can never explain the difference between two inverters at the same plant — by construction they experienced identical weather. It can only explain plant-level and time-level variation.	**STRUCTURAL**	Join Fact_Weather → Dim_Plant, and reach inverters via Dim_Plant → Dim_Inverter. Never build a direct Weather ↔ Generation relationship.
S1	All sheets	No asset register exists anywhere in the dataset. No capacity (kWp), no panel model or count, no inverter make, no location, no commissioning date. Four fact tables, not one dimension table.	Specific yield (kWh/kWp) — the headline metric of the previous project — is impossible, because there is no denominator.	**HIGH**	Accept and substitute something better: Performance Ratio from irradiation (AC output per unit of irradiance)

ID	Where	Finding (measured across all rows)	Why it matters	Severity	Action
					needs no nameplate capacity and is a truer measure of health than kWh/kWp ever was.
S2	All sheets	SOURCE_KEY is an opaque 15-character hash, e.g. '1BY6WEcLGh8j5v7'. 44 distinct values, no two shared between plants.	Unreadable on a chart axis, in a slicer, or in a KPI card. A dashboard labelled with hashes is unusable.	**MEDIUM - fixed**	Fixed — Dim_Inverter assigns readable labels P1-INV01 … P2-INV22 while keeping the hash as the join key. Use InverterLabel everywhere; never SOURCE_KEY.
T1	Whole dataset	The dataset spans 34 days only — 15/05/2020 to 17/06/2020, confirmed as exactly 34 distinct dates on both plants.	No year-on-year, no seasonal profile, no month-over-month trend worth plotting, and SAMEPERIODLASTYEAR returns blank. A scope limit, not a data defect — nothing can fix it.	**CRITICAL - scope**	Accept and pivot the analysis from calendar to time-of-day. With 34 days × 96 intervals this is an excellent intraday dataset — hour-of-day curves, clipping detection, morning/afternoon asymmetry.
X1	Cross-table	Referential integrity: PLANT_ID is single-valued on each sheet (4135001 / 4136001), exactly 22 distinct inverters per plant, no inverter hash appears in both plants. No orphan keys.	Relationships will validate first time with no blank dimension members.	**PASS**	No action.
X2	Cross-table	Consistency test: DAILY_YIELD never exceeds TOTAL_YIELD on any of the 136,476 generation rows. Timestamps are perfectly aligned between generation and weather — both run 15/05 00:00 to 17/06 23:45 on the same 15-minute grid.	The two fact tables can be joined at exact interval level — something the previous dataset could not do because its grids were offset.	**PASS**	No action — and it unlocks interval-level Performance Ratio.
W3	Both weather sheets	IRRADIATION has a non-zero noise floor at night. Hours 00–04: a few rows up to 0.00031. Hour 05: 26% of rows positive up to 0.00284. Hours 19–23 up to 0.00287. Midday (11–13) peaks at 1.04–1.15. Hour 06 is 100% positive (max 0.148, genuine dawn); hour 18 is 93% positive (max 0.105, genuine dusk).	A filter of IRRADIATION > 0 to mean 'daylight' silently admits night noise rows, dragging any 'efficiency while generating' or Performance Ratio average toward zero, and can produce absurd ratios where the denominator is ~0.0001.	**HIGH**	Never use IRRADIATION > 0. Use a noise threshold of 0.005 instead (100× above the noise ceiling, 200× below midday) as a labelled, editable input in Dim_Time. The generating window is 06:00–18:45.

4. Cleaning actions taken

The source had four fact tables and zero dimension tables, so all four dimensions below were synthesised from the facts themselves. Every derived value is a live formula — nothing is typed in.

Ref	Action	Where	What it achieves	Fixes
C1	Synthesised a date dimension	Dim_Date — 34 rows	15/05/2020 – 17/06/2020 with DayName, DayType, DayIndex, MonthName, ISO week and a live GenRowsThatDay count. Reconciliation passed: those 34 daily counts sum to 136,476, exactly matching 68,778 + 67,698 — proof no generation row falls outside the window.	T1
C2	Synthesised a time dimension — the most important new asset	Dim_Time — 96 rows	TimeKey as minutes-since-midnight (0–1425), plus TimeLabel, Hour, Minute, PartOfDay and WindowFlag. 52 of the 96 intervals fall inside the generating window. With only 34 days available, time-of-day replaces the calendar as the primary analytical axis.	T1, W3
C3	Made the daylight thresholds editable inputs	Dim_Time H2:I4	Noise threshold (0.005) and window hours (6 to 18) sit in labelled blue-on-yellow input cells that the WindowFlag formula reads. Change the cell and the whole dimension re-flags.	W3
C4	Synthesised a plant dimension	Dim_Plant — 2 rows	Plant name, weather sensor key, and live counts: 22 inverters each, 68,778 / 67,698 generation rows, 3,182 / 3,259 weather rows against 3,264 expected — giving weather coverage of 97.5% and 99.8%.	W1
C5	Put the DC correction in a labelled input, not a hardcoded formula	Dim_Plant I2:I3	DC_ScaleFactor = 0.1 for Plant 1 and 1.0 for Plant 2, in input cells with the supporting evidence recorded alongside. The DAX column multiplies by RELATED(Dim_Plant[DC_ScaleFactor]), so no magic '/10' anywhere in the model.	G1
C6	Synthesised an inverter dimension with readable labels	Dim_Inverter — 44 rows	P1-INV01 … P2-INV22 now replace the 15-character hashes on every axis, slicer and KPI card. The original SOURCE_KEY hash is retained as the join key, so nothing is lost.	S2
C7	Added per-inverter coverage measurement	Dim_Inverter E:H	LoggedIntervals, ExpectedIntervals, Coverage% and CoverageFlag, all live. Result: 40 inverters flagged 'Good' (≥95%) and 4 flagged 'Poor - under 80%', sitting at 72.2% (2,355 of 3,264).	G4
C8	Added a live efficiency audit that proves the DC fix	Dim_Inverter K:N	RawEfficiency, DC_ScaleFactor, CorrectedEfficiency and DataFlag. Raw efficiency reads 9.8% on all 22 Plant 1 inverters and 97.8% on all 22 Plant 2 inverters; corrected reads 97.8% on both. Stays in the workbook as a permanent check.	G1

Ref	Action	Where	What it achieves	Fixes
C9	Converted all 8 ranges to named Excel Tables	All 8 sheets	Dim_Date 34, Dim_Time 96, Dim_Plant 2, Dim_Inverter 44, Gen_Plant1 68,778, Gen_Plant2 67,698, Weather_Plant1 3,182, Weather_Plant2 3,259 — all row counts verified after creation.	—
C10	Decision: no helper columns on the fact sheets	The 4 fact sheets	DateKey, TimeKey and corrected DC are all DAX calculated columns rather than 136,476 worksheet formulas × 3. Keeps the workbook fast, and calculated columns can still carry relationships.	G1
C11	Decision: integer join keys, never datetime equality	Model design	Joining on a floating-point datetime risks silent mismatches in the 15th decimal place. The facts instead join on DateKey and TimeKey — exact, robust, and it gives you the intraday axis for free.	X2
C12	Decision: excluded TOTAL_YIELD from the model	Fact_Generation	71.6% of Plant 2 values are on the wrong scale, so it cannot be repaired by trimming outliers. All energy figures derive from AC_POWER, which is clean on both plants. Hide the column rather than delete it.	G2
C13	Decision: source sheets left completely untouched	All 4 source sheets	Not one source value was edited, overwritten or deleted. Every fix is either a formula in a new dimension sheet or a DAX column in the model, so the clean is fully reversible and auditable.	—

5. Data model — what to load and what to connect

5a. First: append the plant pairs into single fact tables

Both plants share an identical schema and PLANT_ID is on every row, so they belong in one fact table each. Loading them as four separate tables would force every measure to be written twice and then added together.

#	Step	Detail	Result
1	Append the two generation tables	Click inside Gen_Plant1 → Data → From Table/Range (opens Power Query) → Close & Load To → Only Create Connection. Repeat for Gen_Plant2. Then Data → Get Data → Combine Queries → Append → pick Gen_Plant1 and Gen_Plant2 → name the result Fact_Generation → Close & Load To → Only Create Connection + Add this data to the Data Model.	Fact_Generation = 136,476 rows (68,778 + 67,698). PLANT_ID distinguishes the two plants.
2	Append the two weather tables	Same pattern with Weather_Plant1 and Weather_Plant2 → name the result Fact_Weather.	Fact_Weather = 6,441 rows (3,182 + 3,259).
3	Load the four dimensions	For each of Dim_Date, Dim_Time, Dim_Plant, Dim_Inverter: click inside the table → Power Pivot tab → Add to Data Model. Small tables, so the simple route is fine.	34 + 96 + 2 + 44 rows. Six model tables in total.
⚠	**Do NOT load the four source tables as well**	Gen_Plant1, Gen_Plant2, Weather_Plant1 and Weather_Plant2 must stay as connection-only queries feeding the appends.	Loading both the parts and the appended whole double-counts every single interval.

5b. Then: add these 5 DAX calculated columns

Order matters. The DC correction uses RELATED(), which needs relationships 1 and 4 to exist first. So: build relationships → then add DC_Power_Corrected.

On	Column name	DAX	Why
Gen	DateKey	= INT (Fact_Generation[DATE_TIME])	Integer date. Joins to Dim_Date without any floating-point risk.
Gen	TimeKey	= HOUR (Fact_Generation[DATE_TIME]) * 60 + MINUTE (Fact_Generation[DATE_TIME])	Minutes since midnight, 0–1425. Joins to Dim_Time and gives you the intraday axis that replaces the calendar.
Gen	DC_Power_Corrected	= Fact_Generation[DC_POWER] * RELATED (Dim_Plant[DC_ScaleFactor])	Fixes issue G1. Reads the 0.1 / 1.0 factor from the labelled input cell in Dim_Plant instead of hardcoding a '/10'. Use this column everywhere and never DC_POWER. Add it after relationships 1 and 4 exist.
Weather	DateKey	= INT (Fact_Weather[DATE_TIME])	Same integer-date pattern.
Weather	TimeKey	= HOUR (Fact_Weather[DATE_TIME]) * 60 + MINUTE (Fact_Weather[DATE_TIME])	Puts weather and generation on the same date and time axes — this is what makes interval-level Performance Ratio possible.

5c. The 7 connections to make

All seven are many-to-one with single filter direction (the default). In Power Pivot → Manage → Diagram View, drag from the first column to the second. Every arrow points into a dimension.

#	From (many side)	To (one side)	What it gives you	Type
1	Fact_Generation[SOURCE_KEY]	Dim_Inverter[SOURCE_KEY]	Readable inverter labels, coverage% and the efficiency audit on every visual. Build this one first — the DC correction depends on it.	* → 1
2	Fact_Generation[DateKey]	Dim_Date[DateKey]	Day-level slicing across the 34 days, plus weekday/weekend splits.	* → 1

#	From (many side)	To (one side)	What it gives you	Type
3	Fact_Generation[TimeKey]	Dim_Time[TimeKey]	The most valuable relationship in this model. Hour-of-day curves, clipping detection, morning vs afternoon asymmetry, and the generating-window filter.	* → 1
4	Dim_Inverter[PLANT_ID]	Dim_Plant[PLANT_ID]	Snowflakes the plant onto the inverter, so a plant filter reaches generation. Also what lets RELATED() find DC_ScaleFactor.	* → 1
5	Fact_Weather[PLANT_ID]	Dim_Plant[PLANT_ID]	The only correct way to link weather to anything. Weather is plant-grain, so it attaches to the plant, never to an inverter.	* → 1
6	Fact_Weather[DateKey]	Dim_Date[DateKey]	Puts weather on the same day slicer as generation.	* → 1
7	Fact_Weather[TimeKey]	Dim_Time[TimeKey]	Aligns irradiance and output at the exact same 15-minute slot — the foundation of Performance Ratio.	* → 1

Do not create these four

- (a) Fact_Weather[SOURCE_KEY] → Dim_Inverter[SOURCE_KEY] — the big trap. Both tables have a column literally named SOURCE_KEY, so Power Pivot may even suggest this join. But the weather SOURCE_KEY is a sensor id (HmiyD2TTLFNqkNe / iq8k7ZNt4Mwm3w0) and those two values appear nowhere in Dim_Inverter. The join would silently match nothing.
- (b) Fact_Generation[PLANT_ID] → Dim_Plant[PLANT_ID] — creates a second path to Dim_Plant alongside relationships 1 + 4. Ambiguous. Hide the column and reach plant via Dim_Inverter.
- (c) Fact_Weather ↔ Fact_Generation directly — plant grain vs inverter grain. Many-to-many; it will double-count 22-fold.
- (d) Any bidirectional / 'Both' filter direction — two facts already share three dimensions; 'Both' closes a circular path.

5d. Model settings — easy to skip, each one breaks something

#	Setting	Where	What breaks if you skip it
1	Sort PartOfDay by PartOfDayOrder	Dim_Time → select PartOfDay → Sort by Column → PartOfDayOrder	Without it the axis reads Afternoon, Midday, Morning, Night — alphabetical nonsense. PartOfDayOrder (1–5) exists specifically for this.
2	Sort TimeLabel by TimeKey	Dim_Time → TimeLabel → Sort by Column → TimeKey	Zero-padded 24-hour labels happen to sort correctly by accident, but do not rely on it — set it explicitly.
3	Set DAILY_YIELD default aggregation to MAX	Fact_Generation → DAILY_YIELD → Summarize By → Max	Fixes issue G3. It is a within-day running total, so the default Sum inflates it roughly 96-fold. Setting the default means a careless drag cannot produce a wrong number.
4	Hide the columns that must never be used	Right-click → Hide from Client Tools	Hide: Fact_Generation[DC_POWER] (raw, inflated), Fact_Generation[TOTAL_YIELD] (issue G2), Fact_Generation[PLANT_ID] (use Dim_Plant), both [SOURCE_KEY] columns (hashes — use InverterLabel), and both DATE_TIME columns (use DateKey + TimeKey).
5	Mark Dim_Date as a Date Table	Dim_Date → Design → Mark as Date Table → DateKey	Needed for any DAX date function to work at all. Be realistic though: with 34 days, YTD and SAMEPERIODLASTYEAR return blank. Set it for correctness, not for time intelligence.
6	Verify after loading	Power Pivot → Manage → row counts at the foot of each tab	Fact_Generation must read 136,476 and Fact_Weather 6,441. If generation reads 68,778 or 67,698 the append failed; if it reads 272,952 you loaded the parts and the whole.

6. What this means for the three dashboards

The three perspectives survive unchanged — the consumer is still a plant performance manager asking the same three questions. But the 34-day span kills every calendar-based visual, and irradiance unlocks better replacements.

Dash	Perspective	Dies with this dataset	Replaced by — and these are better
1	Plant & Inverter Performance	Monthly trend, seasonal profile, year-on-year, specific yield (kWh/kWp — no capacity data exists).	Performance Ratio (AC output per unit irradiance) — the metric the previous project wanted and could never build. Plus daily output across the 34 days, inverter league table on PR, and Plant 1 vs Plant 2 head-to-head. PR needs no nameplate capacity, so it covers all 44 inverters with no caveat — unlike the old dataset where the headline KPI reached only 60% of the fleet.
2	Asset Health & Reliability	Almost nothing — this perspective transfers intact and is still the strongest of the three.	Coverage% by inverter (4 inverters already flagged under 80%), inverter efficiency (AC/DC — newly possible, and it caught issue G1), hour-of-day output curve for clipping and shading, and module vs ambient temperature to expose thermal derating. Three of those four could not be built at all on the old dataset.
3	Weather-Normalised Yield	Seasonal comparison, and the whole 'is it the weather or us?' framing over a long horizon.	Transformed from the weakest perspective to arguably the strongest. Real irradiance means output-vs-irradiance scatter with a genuine fitted relationship, PR by time of day, and module-temperature derating curves. The old version rested on temperature as a proxy for sunshine; this one rests on a measured irradiance sensor.

7. D_Analysis — overview KPI engine & master measure list

Everything the dashboards depend on starts here. Build the measures (Step 1), then the 5 overview pivots (Step 2), then wire D_Overview to the KPI strip (Step 3).

7.0 End consumer & the three perspectives (set for this dataset)

Consumer	**The Plant Operations & Performance Engineer**	Accountable for the output and equipment health of the two solar plants and their 44 inverters.
Why	This is an operations dataset, not a reporting one.	15-minute inverter telemetry plus on-site irradiance and module-temperature sensors, over a 34-day window. That is precisely an operational review pack — and 34 days is a normal operations review period, so the short span stops being a weakness and becomes the natural scope.
Ruled out	Four consumers the data cannot serve.	Executive / CFO — no tariff, cost or revenue data anywhere, and 34 days cannot carry a business trend. Sustainability reporting — no emission factors, no consumption baseline, and 34 days is not a reporting period. Asset investment / due diligence — needs nameplate capacity, age and warranty; none exist. Grid operator — no export or demand data. Choosing any of these would force you to invent numbers.

Dash	Perspective	The question it answers	Why it earns a place
1	Plant Output & Performance	"Are the two plants converting the sunlight they actually received into power as well as they should?"	The outcome lens — what the engineer reports upward. Built on Performance Ratio, which is only possible because this dataset has a real irradiance sensor. Covers all 44 inverters with no coverage caveat.
2	Inverter Fleet Health	"Which of the 44 inverters is underperforming — and is it a hardware fault or just missing data?"	The diagnostic lens. Needs no weather at all, so it is immune to every weather caveat. AC/DC conversion efficiency is newly measurable here — and it is what exposed issue G1.
3	Environmental Response & Losses	"How much output are we losing to heat and to time of day, and which part of it is avoidable?"	The physics lens — it stops the engineer chasing a fault that was really a hot afternoon. Measured irradiance and module temperature make this rigorous rather than suggestive.
Chain	**The three chain together**	Dashboard 1 says whether there is a problem. Dashboard 2 says which inverter it is. Dashboard 3 says whether it is the equipment or the weather. No two answer the same question, and none needs data you do not have.	

7.1 Create these 20 measures first

Power Pivot → Manage → Home → Measures → New Measure. The one constant: 0.25 = 15 minutes expressed in hours (15/60). AC_POWER and DC_Power_Corrected are instantaneous kW, so energy = power × interval hours. If the logging interval ever changes, measures 1, 2 and 6 must all change.

#	Measure name	DAX	Purpose / warning
1	Total AC Energy (kWh)	`SUM ( Fact_Generation[AC_POWER] ) * 0.25`	The base energy measure. AC is clean on both plants, unlike TOTAL_YIELD (issue G2). Everything else builds on this.
2	Total DC Energy (kWh)	`SUM ( Fact_Generation[DC_Power_Corrected] ) * 0.25`	Uses the corrected column. Never SUM DC_POWER raw — Plant 1 is inflated 10× (issue G1).
3	Inverter Efficiency %	`DIVIDE ( [Total AC Energy (kWh)], [Total DC Energy (kWh)] )`	Should read ~97.8% on both plants. If it reads ~9.8% the DC_Power_Corrected column is missing or wrong. Doubles as a permanent health check on the model.
4	Peak AC Power (kW)	`MAX ( Fact_Generation[AC_POWER] )`	Highest single-interval output in scope. Used for clipping detection on Dashboard 3.
5	Avg AC While Generating (kW)	`CALCULATE ( AVERAGE ( Fact_Generation[AC_POWER] ), Dim_Time[WindowFlag] = "Generating window" )`	A plain AVERAGE is halved by the ~50% of intervals that are night zeros (issue G5). This restricts to 06:00–18:45 via Dim_Time.
6	Total Irradiation (kWh/m²)	`SUM ( Fact_Weather[IRRADIATION] ) * 0.25`	Total solar energy that actually arrived. The denominator of every performance metric.
7	Avg Irradiance (kW/m²)	`AVERAGE ( Fact_Weather[IRRADIATION] )`	Use on a time-of-day axis. Across all hours it is diluted by night rows and by the noise floor (issue W3).
8	Peak Irradiance (kW/m²)	`MAX ( Fact_Weather[IRRADIATION] )`	Expect ~1.22 on Plant 1 and ~1.10 on Plant 2. Pairs with Peak AC Power to show whether the plant clipped.
9	Estimated Capacity (kW)	`SUM ( Dim_Inverter[EstCapacity_kW] )`	Estimated, not nameplate — there is no capacity data in the source (issue S1). Derived as each inverter's peak observed corrected DC. Totals 31,241 kW for Plant 1 and 30,249 kW for Plant 2 — two matched ~30 MW plants, which is itself a validation. Always label it 'estimated' on a dashboard.
10	Estimated Performance Ratio	`DIVIDE ( [Total AC Energy (kWh)], SUMX ( VALUES ( Dim_Plant[PlantName] ), [Total Irradiation (kWh/m²)] * [Estimated Capacity (kW)] )`	The headline metric — the one the previous dataset could never produce. Note the SUMX: the denominator must be computed per plant and then summed. A plain ratio of sums would multiply Plant 1's irradiance by Plant 2's capacity and be badly wrong whenever both plants are in scope. Dimensionless, so 0.85 = 85%.

#	Measure name	DAX	Purpose / warning
		))	
11	Irradiance-Normalised Yield	DIVIDE ([Total AC Energy (kWh)], SUMX (VALUES (Dim_Plant[PlantName]), [Total Irradiation (kWh/m²)]))	The assumption-free alternative to measure 10 — kWh out per kWh/m² in, with no capacity estimate involved. Not dimensionless, so use it only to compare (inverter vs inverter, hour vs hour), never as an absolute percentage.
12	Logged Intervals	COUNTROWS (Fact_Generation)	How many readings actually exist in scope. Should read 136,476 with no filters.
13	Expected Intervals	COUNTROWS (Dim_Date) * COUNTROWS (Dim_Time) * COUNTROWS (Dim_Inverter)	How many readings should exist. All three dimensions respect slicers, so this stays correct when you filter to one plant or one day. Reads 34 × 96 × 44 = 143,616 unfiltered.
14	Data Coverage %	DIVIDE ([Logged Intervals], [Expected Intervals])	Fixes issue G4. Makes missing data visible instead of silently depressing totals. Should read ~95% overall; four inverters sit at 72%.
15	Inverters Reporting	DISTINCTCOUNT (Fact_Generation[SOURCE_KEY])	How many inverters sent any data in scope. 44 unfiltered; drops if you filter to a single interval where some were offline.
16	Inverters Below 80% Coverage	COUNTROWS (FILTER (VALUES (Dim_Inverter[InverterLabel]), [Data Coverage %] < 0.8))	The action KPI — a count the engineer can act on today. Reads 4 unfiltered. The 0.8 threshold is the one hardcoded business number in the measure set; it is stated here so it is easy to find and change.
17	Avg Ambient Temp (°C)	AVERAGE (Fact_Weather[AMBIENT_TEMPERATURE])	Air temperature. Expect ~25.5°C on Plant 1 and ~28.1°C on Plant 2.
18	Avg Module Temp (°C)	AVERAGE (Fact_Weather[MODULE_TEMPERATURE])	Panel surface temperature — newly available in this dataset. This is what actually drives efficiency loss, not air temperature.
19	Module Temp Rise (°C)	[Avg Module Temp (°C)] - [Avg Ambient Temp (°C)]	How much hotter the panels run than the surrounding air — peaks around midday and is the direct driver of thermal derating. The core insight on Dashboard 3.
20	Coverage Caption	"Covers " & [Inverters Reporting] & " of " & COUNTROWS (ALL (Dim_Inverter)) & " inverters, " & COUNTROWS (ALL (Dim_Date)) & " days" & " | Capacity is ESTIMATED - no nameplate data"	A text measure. Drop it into a card on every dashboard so the scope and the capacity caveat travel with the numbers instead of living only in this sheet.